Workshop Part 1: Foundation

🎯 Learning Objectives

By completing this workshop, you will:

- ☐ Understand the architecture of this pattern
- ☐ Set up the development environment
- ☐ Deploy infrastructure
- ☐ Configure basic services

⏱️ Estimated Duration: 3-4 hours

📋 Prerequisites

- Azure subscription
- Basic understanding of Kubernetes
- Docker knowledge
- CLI tools installed (az, kubectl, helm)

🚀 Steps

Step 1: Environment Setup (30 min)

1. Clone this repository
2. Set up Azure CLI
3. Create resource group
4. Configure kubectl access

```
# Create resource group
az group create --name myapp-rg --location eastus

# Create AKS cluster (if not using existing)
az aks create --resource-group myapp-rg --name myapp-aks --node-count 3

# Get credentials
az aks get-credentials --resource-group myapp-rg --name myapp-aks
```

Step 2: Infrastructure Deployment (60 min)

1. Review architecture diagram in `architecture/system-overview.png`
2. Deploy infrastructure using Bicep/Terraform
3. Verify all resources are created

```
# Deploy using Bicep
az deployment group create \
  --resource-group myapp-rg \
  --template-file deployment/bicep/main.bicep

# Verify deployment
```

```
kubectl get nodes
kubectl get pods -A
```

Step 3: Deploy Core Services (60 min)

1. Apply Kubernetes manifests
2. Verify services are running
3. Check health endpoints

```
# Deploy services
kubectl apply -f deployment/k8s/

# Verify
kubectl get svc
kubectl get deployment
```

Step 4: Configure Monitoring (30 min)

1. Deploy monitoring stack
2. Set up dashboards
3. Create alert rules

```
# Check monitoring
kubectl get pods -n monitoring
```

✅ Validation Checklist

- ☐ Cluster is running
- ☐ All pods are in Running state
- ☐ Services have IP addresses
- ☐ Health endpoints respond
- ☐ Logs are being collected
- ☐ Metrics are being collected

📌 Key Concepts

- **Namespaces**: Logical isolation
- **Deployments**: Manages pod replicas
- **Services**: Expose pods to network
- **Ingress**: Route traffic to services
- **ConfigMaps**: Non-secret configuration
- **Secrets**: Sensitive data management

🔗 Related Resources

- Architecture Overview
- Production Hardening
- Troubleshooting

🆘 Common Issues

Issue: Pods not starting

- Check pod logs: `kubectl logs <pod-name>`
- Check resource limits: `kubectl describe pod <pod-name>`
- Verify secrets exist: `kubectl get secrets`

Issue: Service not accessible

- Check service endpoints: `kubectl get endpoints`
- Verify ingress configuration: `kubectl get ingress`
- Test connectivity: `kubectl exec -it <pod> -- curl <service>`

Ready for Part 2? Proceed to part2.md